
the Worldwide Sustainability timeline

About the Worldwide Sustainability Timeline

Concerns about the environment and development are not new. The ongoing global dialogue is centred around the strategies needed to address the inter-related challenges of building healthy societies, economies, and environments. This dialogue has roots in the gradual merging of the environmental movement and the post World War II international development community. Over the past 55 years, optimism about a modern technological utopia has been replaced by a realistic understanding of the forces contributing to the world's problems. Many consider 1962 as the seminal year where people began to understand how closely linked the environment, economy and social well-being truly are. Since then, many milestones have marked the journey towards sustainable development as featured in this timeline.

1960s



1962

ENVIRONMENT

1962 – Rachel Carson publishes "Silent Spring". This book brings together research on toxicology, ecology and epidemiology. It suggests that agricultural pesticides are building to catastrophic levels. This is linked to damage to animal species and human health. It shatters the assumption that the environment has an infinite capacity to absorb pollutants.

1967

POLLUTION

1967 – Torrey Canyon oil tanker runs aground and spills 117,000 tons of oil into the North Sea near the United Kingdom. The massive local pollution prompts legal changes to make ship owners liable for all spills.

1970s

1968

SINGAPORE

1968 – Launch of Keep Singapore Clean Campaign as the first environmental awareness campaign aimed at encouraging Singaporeans to create and maintain a clean environment. Its success sparks off a series of similar campaigns over the next decade.





GOVERNANCE

1968 – Worldwide Meeting of Environmental Experts takes place for the first time in the Intergovernmental Conference for Rational Use and Conservation of Biosphere (UNESCO). They convene in Paris to discuss global environmental problems, including pollution, resource loss and wetlands destruction. This provided a forum for early discussions of the concept of ecologically sustainable development.

1969

SINGAPORE

1969 – Environmental Public Health Act Enacted, providing a strong legal basis for the administration of public health practices such as public cleansing, mosquito and vector control, as well as food hygiene.

1970

GOVERNANCE

1970 – First Earth Day held as a national teach-in on the environment. An estimated 20 million people participate in peaceful demonstrations across the USA to protest against environmental abuses. It sparks the creation of landmark environmental laws including the Endangered Species Act and the Safe Drinking Water Act.

GOVERNANCE

1972 – United Nations Conference on Human Environment held in Stockholm is attended by participants from 114 countries. Only one participant is an environment minister, as most countries do not yet have environmental agencies. The conference is rooted in the regional pollution and acid rain problems of northern Europe. This eco-agenda is opposed by the Group of 77 and the Eastern bloc. Nevertheless, it is the first international recognition of environmental issues. The concept of sustainable development is cohesively argued to present a satisfactory resolution to the environment versus development dilemma. The conference leads to the establishment of numerous national environmental protection agencies and the United Nations Environment Programme (UNEP).

SINGAPORE

1972 – Ministry of the Environment (ENV) Formed to deal with all aspects of pollution, drainage and environmental health. ENV spearheads many efforts to contribute significantly to Singapore's development as a clean and green city.

1972

ECONOMY

1972 – Club of Rome Publishes "Limits to Growth". The report is extremely controversial.It predicts that if growth is not slowed, the earth's limits will be reached in 100 years, at current rates of population growth, resource depletion, and pollution generation. Northern countries criticise the report for not including technological solutions. Southern countries are incensed as the report advocates the abandonment of economic development. The ensuing debate heightens awareness of the interconnections between several well-known global problems.



ECONOMY

1972 – OPEC Oil Crisis Fuels Limits to Growth Debate. Arab country members of the Organisation of Petroleum Exporting Countries (OPEC) reduce oil exports to Europe. They initiate an oil embargo against the United States for its support of Israel in a war with Egypt and Syria.

1973



ENVIRONMENT

1973 – Chipko Movement is Born in India in response to deforestation and environmental degradation. The actions of the women in the community influence both forestry and women's participation in environmental issues. The movement was initiated by women living in Himalayan villages in Northern India. They want to protect trees from commercial logging as it had begun to cause severe deforestation, soil erosion and flooding.

1974



OZONE LAYER

1974 – Seminal Work on CFCs Published in 'Nature' Magazine. Chemists Sherwood Rowland and Mario Molina publish their landmark findings that chlorofluorocarbons (CFCs) can destroy ozone molecules, and erode the Earth's protective ozone layer. They calculate that if human use of CFCs continues at an unaltered rate, the ozone layer would be greatly depleted after some decades.

1975

ENVIRONMENT

1975 – Worldwatch Institute is founded in the USA by environmentalist Lester Brown. This independent research organisation promotes an environmentally sustainable and socially just society. The institute also raises public awareness on global environmental threats, to the point that supports effective policy responses.

BIODIVERSITY

1975 – Convention on International Trade in Endangered Species of Flora and Fauna (CITES) comes into effect. The treaty has a broad mandate of restricting trade in roughly 5,000 animal species and 25,000 plant species threatened with extinction. However, inadequate enforcement allows a billion dollar black market in wildlife trade to flourish.

1980s

ENVIRONMENT

1977 – Blocking of Dam Project Indigenous protestors in the Philippines force the World Bank to withdraw its financial backing for the construction of four large dams along the Chico River. The effort to block the projects energises a global movement to protect rivers by resisting dam building.

1977

SINGAPORE

1978 – Waste Incineration introduced drastically reduces waste volume by up to 90%. It also saves land required for landfills.

1978





1979

GOVERNANCE

1979 – Adoption of the Convention on Long-Range Trans boundary Air Pollution. The convention helps combat acid rain and regulates pollution traveling across national borders. Later protocols regulate emissions of nitrogen oxides, sulphur, heavy metals, persistent organic pollutants and several other pollutants.



1982

GOVERNANCE

1982 – Adoption of the UN Convention on the Law of the Sea. It establishes material rules concerning environmental standards on ocean conservation. It has enforcement provisions dealing with pollution of the marine environment, as well as protection and restoration of species populations.

CLIMATE CHANGE

1983 – Greenhouse Gases linked to Global Warming. The US Environmental Protection Agency and the US National Academy of Sciences release reports concluding that the build-up of carbon dioxide and other "greenhouse gases" in the earth's atmosphere will likely lead to global warming.

1983





1984

POLLUTION

1984 – Bhopal Chemical Leak leaves 10,000 dead and 300,000 injured when Union Carbide's pesticide plant in India sends 40 tonnes of methyl isocyanate gas into the air. The cloud of poison leaks into the surrounding city of one million.

OZONE LAYER

1985 – Antarctic Ozone Hole discovered by British and American scientists. Data from a British Antarctic Survey show that January ozone levels had dropped 10% below that of the previous year.



1985

CLIMATE CHANGE

1985 – Build-up of Greenhouse Gases. A meeting called by the World Meteorological Society, UNEP and International Council of Scientific Unions (ICSU) reports on the build-up of carbon dioxide and other greenhouse gases in the atmosphere. They predict global warming.



POLLUTION

1986 – Accident at Nuclear Station in Chernobyl generates a massive toxic radioactive explosion. One of the four reactors at the Soviet Union's Chernobyl nuclear power plant explodes and completely melts down. The explosion sends radioactive particles as far away as Western Europe, exposing hundreds of thousands of people to high levels of radiation.

1986



GOVERNANCE

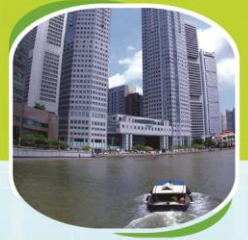
1987 – "Our Common Future" (Brundtland Report) Published by the World Commission on Environment and Development concludes that preserving the environment, addressing global inequities, and fighting poverty could fuel, not hinder, economic growth by promoting sustainable development. It also popularises the term "sustainable development".

OZONE LAYER

1987 – The Montreal Protocol on Substances that Deplete the Ozone Layer is Adopted to support the phasing out of production of the number of ozone-depleting chemicals.

SINGAPORE

1987 – Completion of the Singapore River Clean-up. The 10-year long clean-up saw more than 26,000 families who lived without proper sanitation resettled; 2,800 backyard trades and industries moved into industrial estates; pig and duck farms located near the river phased out; as well as street hawkers and vegetable wholesalers moved into proper hawker centres and the Pasir Panjang wholesale market respectively. The river is transformed into an attractive lifestyle destination, with fishes and other aquatic life returning to it.



1987



TOXICS

1989 – Adoption of The Basel Convention. The convention controls the movement of hazardous waste across international borders. It prevents "toxic traders" from shipping hazardous waste from industrial to developing countries.

1989

1990s

1990



SINGAPORE

1990 – Launch of Clean & Green Week. It marks a new approach to environmental education, where environmental programmes are provided with more structure. The community is encouraged to care for the environment. Its annual appearance has raised awareness amongst Singaporeans of their social responsibility towards the environment.

1992

GOVERNANCE

1992 – Earth Summit. Most countries and 117 heads of state participate in the groundbreaking Earth Summit (UN Conference on Environment and Development), in Rio de Janeiro, Brazil. Participants adopt Agenda 21, a voluminous blueprint for sustainable development that calls for improving the quality of life on earth. Other agreements reached include the Convention on Biological Diversity, the Framework Convention on Climate Change, the Rio Declaration, and the non-binding Forest Principles. Concurrent NGO Global Forum publishes alternative treaties.



CLIMATE CHANGE

1992 – The Convention on Climate Change sets non-binding carbon dioxide reduction goals for industrial countries.



BIODIVERSITY

1992 – The Convention on Biological Diversity mandates that countries formulate strategies to protect biodiversity, and that industrial countries help implement these strategies in developing countries.



1994

BIODIVERSITY

1994 – IUCN Publishes a Revised 'Red List' of Endangered and Threatened Species. It creates a world standard for gauging threats to biodiversity. Later versions report that one in four mammal species and one in eight bird species face a high risk of extinction.



POPULATION

1994 – The Conference on Population and Development in Cairo, Egypt sees the setting up of a decades-long plan to stabilise and reduce population growth. It has delegates from 183 countries. The plan emphasises the importance of women's education and access to reproductive health care.

GOVERNANCE

1994 – North American Free Trade Agreement (NAFTA) Enters into Force. The side agreement, called the North American Agreement on Environmental Cooperation, establishes the Commission for Environmental Cooperation.

1995



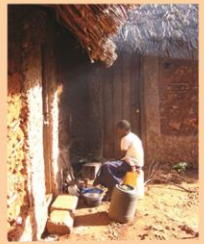
SINGAPORE

1995 – Singapore Enjoys Modern Sanitation Islandwide with 430 million m³ of sewage treated a year by six sewage treatment works, 133 pumping installations, 2,507 km of sewers and an industrial water works. The Sewerage Masterplan was first formulated in 1972, kicking off the development of Singapore's sewerage infrastructure.

CLIMATE CHANGE

1995 – The Intergovernmental Panel on Climate Change (IPCC), a group of prominent climate scientists assembled by the UN in 1988, releases a report concluding that "the balance of evidence suggests that there is a discernible human influence on global climate".





GOVERNANCE

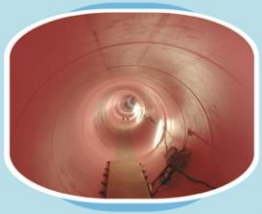
1995 – World Trade Organisation (WTO) Established. Formal recognition of trade, environment and development linkages.

GOVERNANCE

1996 – ISO 14001 Formally Adopted as a voluntary international standard for corporate environmental management systems.

SINGAPORE

1998 – Launch of Deep Tunnel Sewerage System (DTSS) as an innovative and efficient sewerage system that is a significant step towards Singapore's long term sustainability.



GOVERNANCE

1999 – 3rd WTO Ministerial Conference held in Washington sees thousands of demonstrators take to the streets. They are protesting the negative effects of globalisation and the growth of global corporations. Coupled with deep conflicts amongst the delegates, the negotiations are scuttled. This becomes the first of many anti-globalisation protests, signaling a new era of confrontation between disaffected stakeholders and those in power. The protests spotlight the environmental and social shortcomings of the WTO.



GOVERNANCE

1999 – Launch of the First Global Sustainability Index tracking leading corporate sustainability practices worldwide. Called the Dow Jones Sustainability Group Indexes, this tool provides a bridge between companies implementing sustainability principles and investors looking for trustworthy information to guide sustainability focused investment decisions.



WATER

2000 – The 2nd World Water Forum and Ministerial Conference is attended by 5,700 participants and 120 ministers from all parts of the world. It results in the Declaration of The Hague on Water Security in the 21st Century and the release of the 'World Water Vision' for the sustainable use and management of water resources.

CLIMATE CHANGE

2001 – The Marrakech Accords Finalise How the Kyoto Protocol Should Work. Agreed to at COP-7, it lays the groundwork for ratification by different countries and marks the conclusion of a process begun at COP-4 in Buenos Aires.

1996 | 1997 | 1998 | 1999 | 2000s | 2000 | 2001

CLIMATE CHANGE

1997 – Signing of the Kyoto Protocol by Delegates to the UN Framework Convention on Climate Change Third Conference of the Parties (COP-3). This document sets goals for greenhouse gas emission reduction. It establishes emissions trading in developed countries and the clean development mechanism for developing countries. It strengthens the 1992 Climate Change Convention by mandating that industrial countries cut carbon dioxide emissions by 6% to 8% from 1990 levels by 2012. However, the protocol's controversial emission-trading scheme, as well as debates over the role of developing countries, cloud its future.

OZONE LAYER

1998 – Ozone hole over Antarctica grows to 25 million km². The previous record set in 1993 was 3 million km².



SINGAPORE

1999 – Singapore's 1st Offshore Landfill at Semakau Begins Operations. To create an offshore landfill, a 7-km perimeter rock bund was built to enclose the sea between two islands, Pulau Semakau and Pulau Sakeng. The 350-hectare landfill has a capacity of 63 million m³, targeted to last till 2045. Efforts were also made to protect the marine ecosystem and mangroves near the landfill.



TOXICS

2000 – The Treaty on Persistent Organic Pollutants (POPs) requires the complete phasing out of nine persistent, highly toxic pesticides. It also limits the use of several chemicals including dioxins, furans and PCBs.



SINGAPORE

2001 – PUB Reconstituted as the Single Water Authority in Singapore to close the water loop. PUB ensures that Singapore's water supply is sustainable and well-managed by adopting new technologies, improving operations, and facilitating public-private partnerships.





SINGAPORE

2001 – Launch of National Recycling Programme. The rate of participation by households in this voluntary, door-to-door collection programme has been steadily increasing.

2002

CLIMATE CHANGE

2002 – European Union Ratifies the Kyoto Protocol, bringing industrial countries closer to binding reductions in greenhouse gas emissions.

GOVERNANCE

2002 – World Summit on Sustainable Development held in Johannesburg, South Africa. World governments, concerned citizens, UN agencies, multilateral financial institutions and other major groups participate in and assess global change since the United Nations Conference on Environment and Development (UNCED) in 1992. Some 104 world leaders and thousands of delegates agree on a limited plan to reduce poverty and protect the environment.



SINGAPORE

2002 – Formation of the National Environment Agency (NEA) under the Ministry of the Environment and Water Resources (MEWR). NEA's focus is on the implementation of environmental policies. The NEA divisions of Environmental Protection, Environmental Public Health, and Meteorological Services will work together to ensure a quality environment for Singaporeans.

SINGAPORE

2002 – Launch of Singapore Green Plan 2012 charts out Singapore's roadmap towards sustainable development for the next 10 years.





SINGAPORE

2002 – Singapore ratifies the ASEAN Haze Agreement. It is the first legally-binding ASEAN regional accord, signifying Singapore's commitment towards tackling the problem of forest fires and haze.

2003



SINGAPORE

2003 – Launch of NEWater making use of advanced membrane technologies to treat water. In the long run, it ensures greater resilience in Singapore's water supply.

CLIMATE CHANGE

2003 – Europe Adopts 1st Climate Emissions Trading Law, giving carbon dioxide a market value across the EU when trading begins in 2005.



2004



POLLUTION

2004 – First-ever Pollution Register. Issued by the European Union, the register contains a wealth of data on industrial emissions and represents a landmark event in public provision of environmental information.



SINGAPORE

2004 – Semakau Landfill Opens for Recreational Activities. The world's first man-made offshore landfill exists alongside a thriving ecosystem. During construction, efforts were taken to preserve its original biodiversity. Semakau is now open to nature lovers for bird watching, intertidal walks, sports fishing, star gazing and camping.



SINGAPORE

2004 – Singapore's First Desalination Plant. Desalination is one of Singapore's "Four Taps Strategy" to provide a diversified, sustainable supply of water. Recent technological advancements have turned desalination technology into an affordable option for our island-state.



SINGAPORE

2004 – Renaming of MEWR. Ministry of the Environment is renamed as Ministry of the Environment and Water Resources (MEWR) to reflect its significantly expanded role. MEWR's mission is to deliver and sustain a clean and healthy environment as well as water resources for Singapore, through its agencies NEA and PUB.

SINGAPORE

2005 – Singapore Accedes to Various Environmental Conventions. This includes the Stockholm Convention on Persistent Organic Pollutants and the Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade.

GOVERNANCE

2005 – Kyoto Protocol Enters into Force, legally binding developed countries to goals for greenhouse gas emission reductions. It also establishes the Clean Development Mechanism for developing countries.

SINGAPORE

2006 – Singapore Accedes to the Kyoto Protocol of the United Nations Framework Convention on Climate Change.

SINGAPORE

2006 – Singapore Joins the Renewable Energy and Energy Efficiency Partnership. This global initiative promotes policy action on clean energy markets as well as financing for sustainable energy projects.

CLIMATE CHANGE

2007 – Public Attention on Climate Change Increases with 'An Inconvenient Truth', an award-winning documentary by former US Vice President Al Gore. The IPCC's alarming forecast on the planet's health makes headlines. The IPCC and Al Gore share the Nobel Peace Prize.



10% ENERGY CHALLENGE

SINGAPORE

2008 – 10% Energy Challenge. NEA launches a first-of-its-kind national campaign to promote energy conservation habits amongst households. These habits include switching off at mains, using energy efficient appliances and setting air conditioning thermostats to 24°C or higher.

2005

SINGAPORE

2005 – Energy Efficiency Improvement Assistance Scheme (EASe) is a co-funding scheme by NEA to encourage companies in the manufacturing and building sectors to appraise their energy consumption. They can then identify areas for energy efficiency improvements.



2006

CLIMATE CHANGE

2006 – The Stern Report makes a convincing economic case that the cost of inaction on climate change will be up to 20 times greater than the measures required to address the issue today.



2007

SINGAPORE

2007 – Singapore Accedes to the 1999 Beijing Amendment to the Montreal Protocol on substances that deplete the ozone layer.

SINGAPORE

2007 – 'Public Sector Taking the Lead in Environmental Sustainability' Initiative spells out the thrust for Singapore's public sector to demonstrate the environmental and economic benefits of efficient energy and resource use. It sets an example for the private sector to follow. Public sector buildings will be energy-audited and improvements made to meet energy efficiency targets.



ENERGY

2008 – Food, Fuel & Financial Crises. Global food prices increase 43% in one year; growing worldwide demand sends energy prices soaring; the collapse of mortgage lending in the US sends the world into recession.



SINGAPORE PACKAGING AGREEMENT

SINGAPORE

2008 – Inaugural 3R Packaging Awards give recognition to signatories of he Singapore Packaging Agreement for their notable contributions toward sustainable packaging practices.

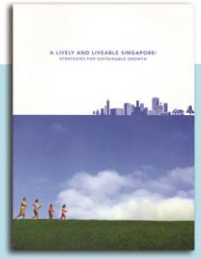


SINGAPORE

2008 – Opening of Marina Barrage. Marina Barrage is a dam built across the mouth of the Marina Channel to separate seawater from freshwater, forming the Marina Reservoir in the heart of the city. Considered an engineering marvel, this project serves to create a new source of water supply, act as a tidal barrier for flood control, and become a lifestyle attraction and water sports hub.

FOSSIL FUELS

2009 – G20 Promises to Phase Out Fossil Fuel Subsidies. Experts estimate that annual subsidies could amount to US$500 billion or 1% of the world's GDP.



SINGAPORE

2009 – MEWR and the Ministry of National Development publish 'A Lively & Livable Singapore: Strategies for Sustainable Growth', referred to as the '2030 Sustainability Blueprint'. It documents the findings and recommendations of the Inter-Ministerial Committee on Sustainable Development, after consultations with the public as well as business and community leaders.



SINGAPORE

2010 – Opening of Singapore's Largest NEWater Plant: Sembcorp NEWater Plant. Housed on top of the Changi Water Reclamation Plant, the innovative plant-on-plant design facilitates large-scale water recycling in land-scarce Singapore. The plant has a total capacity of 228,000 cubic metres of NEWater per day. NEWater now meets 30% of Singapore's total water demand.

2008

SINGAPORE

2008 – Mandatory Energy Labelling Kicks In under the Environmental Protection and Management Act where registrable goods such as air conditioners, refrigerators and clothes dryers must carry energy labels.



GOVERNANCE

2008 – Green Economy Goes Mainstream. National governments invest a portion of their economic stimulus in environmental actions. A low-carbon economy and green growth become new objectives for the future economy.



2009



CLIMATE CHANGE

2009 – Copenhagen Climate Negotiations are seen as critical to measuring whether climate mitigation actions are meeting the global reductions that science demands. The outcomes of the Copenhagen negotiations are unclear: the process may be in trouble but the Copenhagen Accord itself may be a breakthrough in terms of engaging developing countries.



SINGAPORE

2009 – Completion of Deep Tunnel Sewerage System (DTSS) & Changi Water Reclamation Plant (WRP). Through both projects, PUB will collect, treat, reclaim and dispose of Singapore's used water. Running 20 m to 55 m underground, the mammoth DTSS conveys used water to a centralised plant in Changi. The state-of-the-art Changi WRP is capable of treating 800,000 cubic metres (320 Olympic sized pools) of used water a day.

2010

BIODIVERSITY

2010 – COP-10 to the Convention on Biological Diversity in Nagoya, Japan agrees to a 20-point strategy to be implemented by governments in the next 10 years. It will help tackle the mass extinction of species and the loss of vital habitats around the world. By 2020, governments agree to boost the area of protected land in the world to 17%, and strive for marine protected areas covering10% of our oceans.



SINGAPORE

2010 – EENP (Energy Efficiency National Partnership) Learning Network allows the industrial sector to learn and share energy efficient technologies and best practices. Capacity building is done through learning platforms such as seminars, forums, conferences, technical workshops, learning journeys, site visits and roundtable discussions.

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SINGAPORE

2011 – Singapore Exchange (SGX) Launches the 'Guide to Sustainability Reporting for Listed Companies'. It sets out the broad principles for listed companies to formulate a sustainability reporting framework. Investors and shareholders are increasingly recognising the relevance of sustainability or non-financial reporting to companies' risk management capabilities, after a string of corporate missteps in the past few years.



CLIMATE CHANGE

2011 – COP-17 to the UN Framework Convention on Climate Change in Durban delivers a breakthrough on the international community's response to climate change. The negotiations advance the implementation of the Convention and the Kyoto Protocol, the Bali Action Plan, and the Cancun Agreements. The outcomes include a decision to adopt a universal legal agreement on climate change no later than 2015. However, scientists and environmental groups warn that the deal is not sufficient to avoid global warming beyond 2°C and more urgent action is needed.

SINGAPORE

2012 – MEWR's Energy Conservation Act in FY2013 will make it mandatory for large users in the industry and transport sectors to practice energy management. It will require the appointment of energy managers,reporting of energy use and submission of energy efficiency improvement plans. Government assistance will be given to improve energy efficiency, build up energy management capabilities and invest in new technologies.



SINGAPORE

2012 – Re-opening of Kallang River at Bishan Park. Marking a local first, the concrete canal at Bishan Park is converted into a natural meandering river under PUB's Active, Beautiful, Clean Waters Programme. Designed based on a floodplain concept, bioengineering techniques using natural materials help stabilise the slopes, prevent soil erosion, and form natural habitats that enrich the park's biodiversity.



2011

2012

SINGAPORE

2011 – Completion of Punggol & Serangoon Reservoirs, Singapore's 16th and 17th reservoirs respectively. Created by the damming of Punggol and Serangoon rivers, both reservoirs collect rainwater from Punggol, Sengkang, as well as parts of Hougang and Ang Mo Kio. They also bring waterfront living into the heartlands. With the additions, Singapore's water catchment areas now comprise two-thirds of the island.



GOVERNANCE

2012 – UN Conference on Sustainable Development (Rio+20) takes place in Rio de Janeiro, Brazil, presenting a historic opportunity to define pathways that can reduce poverty, promote decent jobs, clean energy and a more sustainable and fair use of resources; address environmental destruction as well as build a bridge to the future. It occurs 20 years after the 1992 Earth Summit in Rio, where countries adopted Agenda 21, a blueprint to rethink economic growth, advance social equity and ensure environmental protection.

RIO+20 United Nations Conference on Sustainable Development

SINGAPORE

2012 – Inaugural CleanEnviro Summit Singapore (CESS) is global platform for thought leaders, government officials and practitioners to identify and develop practical solutions to the environmental challenges of tomorrow's cities. It is held in conjunction with WasteMET Asia, World Cities Summit and Singapore International Water Week.

References:
www.worldwatch.org
www.iisd.org
www.sxc.hu



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